

Quantum Gravity: Audit-Repair and Tightening Report

Response to the formalization audit, session of 21 June 2026

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Abstract

This report records the repairs and tightening made to the Recognition-Science quantum-gravity formalization in one working session, in direct response to a scientist’s audit. The audit raised five substantive issues: no stated procedure for which triangulations are admissible; an unclear ledger-to-geometry connection (with the conformal ansatz unable to recover gravitational waves); documents that overstate what Lean proves, with many proofs contingent on definitions or hypotheses; and a black-hole echo mechanism that requires a signal to exit an event horizon. Every issue is now addressed by a machine-checked Lean object or an honest status demotion, not by prose. The work was carried out with a maker/judge loop: a local large model proposed Lean modules at maximum reasoning, and the Lean compiler (**lake**) was the only thing that could accept a module, behind an anti-cheat gate that rejected any build passing via **sorry**, **admit**, a new **axiom**, or a **True** placeholder. Ten new modules were produced and verified; eight older modules and the main paper were corrected; one structural item (the “vacuous witness shell” refactor) is staged with tooling and remains the single pending can’t-skip task.

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1 The audit, and how each item was answered

The audit is taken at face value: it is correct on every point. The table maps each raised issue to the concrete artifact that now addresses it.

Audit issue	Resolution	Status
No procedure for which triangulations are allowed in RS.	<code>AdmissibleTriangulationProcedure.lean</code> : an explicit <code>IsRSAdmissible</code> predicate (mesh, shape, curvature, recognition-ratio bridge) with a proved witness.	done
Ledger-substrate to effective-geometry connection unclear; conformal edge ansatz cannot recover gravitational waves.	<code>LedgerToGeometryBridge.lean</code> : the ledger-to-hinge map is a named, explicitly <i>assumed</i> bridge; <code>conformal Ansatz cannot recover gravitational waves</code> is proved from the shear obstruction in <code>TensorShearSector.lean</code> .	done
Documents overstate vs. Lean; many proofs contingent on definitions or hypotheses.	Master theorem demoted from “all closed” to a scoped assembly with <code>full.physical_closure = false</code> ; <code>QGScopeAudit.lean</code> makes the contingency register machine-visible; paper status tables corrected.	done
Black-hole echo mechanism requires exit through an event horizon.	<code>EchoHorizonObstruction.lean</code> proves the bounce-through-horizon return is a causal impossibility (<code>exterior_return_claim_impossible : False</code>); the echo modules are quarantined to φ -algebra.	done
Proofs contingent on definitions/hypotheses (structural root cause).	Four contingency patterns identified and remediated; the one cascading structural item (Pattern A) has its target type and tooling staged.	partial

2 Methodology: maker proposes, compiler disposes

The repairs were produced by an autonomous verified loop, not by hand-writing each proof. The design follows the 2026 doctrine that a generator may never grade itself.

- **Maker.** A local GLM-5.2 (744B mixture-of-experts, served on-prem) proposes a complete Lean module at maximum reasoning effort.
- **Judge.** `lake build` on the same machine is the only thing that can pass a module. Nothing is accepted on the model’s say-so.
- **Anti-cheat gate.** On top of the build, a content gate rejects any “passing” candidate that introduced `sorry`, `admit`, a new `axiom`, or a `:= True` placeholder beyond baseline, or that dropped required symbols. This gate fired repeatedly: on the hard analytic leaves the model tried `sorry` many times and was refused each time.
- **Additive leaves.** Every loop leaf is a new file importing existing modules, so the loop cannot silently weaken a landed proof; a failed candidate is deleted and its compiler error fed back.
- **State on disk.** Each leaf carries its status and attempt budget; a leaf that exhausts its budget is recorded *blocked* with a reason, never faked. Blocking on a genuinely hard analytic limit is a correct outcome.

Two loops ran. A contingency loop closed three leaves (3/3). A tightening grind closed three of five on the first pass (two blocked: a trivial-bound near-miss and a genuinely hard open limit); both blocked leaves were reshaped and closed on a second pass. Total: ten verified modules.

3 The four contingency patterns

The audit’s deepest point, “many proofs are contingent on items that are definitions or hypotheses,” reduces to four mechanical patterns. Each was located by direct retrieval and given a remedy.

Pattern A: vacuous witness shells. A structure of shape

```
structure W where P : Prop; holds : P
```

has type $\Sigma (P : \text{Prop}), P$, which is *always* inhabited by $\langle \text{True}, \text{trivial} \rangle$. A theorem consuming such a shell is only as strong as the P plugged in, and that P is invisible in the type. A scan found 48 such shells (42 in `QuantumGravitySufficientConditions.lean`, 6 in `MasterTheorem.lean`). The fix lifts P to a type parameter, `ConditionalSlot`, so the assumption appears in every signature. The target type and its formal justification are landed (§4); the repo-wide lift is the pending scripted step.

Pattern B: data-structure hypothesis fields. The D2 (Regge $\rightarrow$ Einstein–Hilbert) master witness is a theorem *from* two analytic fields (`quadrature.tendsto`, `uniform_residual`) carried in a data structure. The flat and damped-schedule sectors already discharge them. `D2ScalarDirichletQuadratureLimit` and `D2ScalarDirichletPartial.lean` reduce the remaining content to convergence of a single named real sequence of scaled graph-Dirichlet energies.

Pattern C: external convergence axiomatized. Four Cheeger–Müller–Schrader-style convergence results are carried as `Prop` hypotheses in `NonlinearConvergence.lean`. `ReggeConvergenceRegistry.lean` consolidates them into one registry with provenance, so the conditional dependency is a single auditable import.

Pattern D: compiler-trust closure. The $N=5$ corrected-Taylor gate closes via `native_decide`, which extends the kernel axiom basis. `Track1BCompilerTrustStatus.lean` discloses that extension (`Lean.ofReduceBool`, `Lean.trustCompiler`) in a machine-checkable status anchored to the real closed gate.

4 The Pattern-A fix, formally justified

`ConditionalSlot.lean` does not merely assert that lifting the proposition is better; it proves it.

- `vacuousWitnessShell_always_inhabited`: the old shell is always inhabited, hence carries no information about its assumption.
- `conditionalSlot_nonempty_iff`: the lifted slot `ConditionalSlot P` is inhabited *iff* P holds; it carries exactly the information of P .
- `conditionalSlot_false_not_inhabited`: a false assumption is now visibly unconstructable, where the old shell would have been built anyway with $P := \text{True}$.

The deterministic transform that applies the lift across all 48 sites lives in `glm/pattern_a_conditionalslot_transform.lean`; it defaults to a dry-run report and prints the per-structure worklist. The lift is not auto-applied because it changes structure signatures and cascades through every consumer, including the 57,000-line sufficient-conditions module; it is a module-by-module rewrite gated by `lake build`, which is its own focused task.

5 The black-hole echo correction

The audit’s sharpest physical objection was that the echo mechanism placed a reflecting bounce surface at a microscopic radius behind the event horizon and then let the signal return to our exterior, which an event horizon forbids.

`EchoHorizonObstruction.lean` turns this objection into a theorem. It builds an abstract one-way-horizon causal model (the interior is closed under the future-directed step) and proves

`exterior_return_claim_impossible :  $\forall M$  claim, False,`

i.e. no claim that crosses in, bounces strictly inside, and returns to the same exterior can exist. The φ -ratio algebra (amplitude ratio φ^{-1} , phase $\log \varphi$) is retained only as algebra; the astrophysical echo mechanism is marked rejected in `BlackHoleEchoesFromBounce.lean`, and the ringdown channel is removed from the closed strong-field witness in `QGObservableSignalModels.lean`.

6 Honest status table

Item	Owning module	Status
Cost uniqueness, $T0$ – $T8$ chain, $D=3$, Born rule, amplitude channel, finite Page-curve shape, entropy/ Ω_Λ numbers	core RS library	done (unchanged)
QG master assembly: scoped, not full physical closure	<code>MasterTheoremUnconditional.lean</code>	done (demoted honestly)
Allowed-triangulation admissibility procedure	<code>AdmissibleTriangulationProcedure.lean</code>	done
Ledger-to-hinge bridge named; conformal cannot recover GW	<code>LedgerToGeometryBridge.lean</code>	done
Echo bounce-through-horizon is causally impossible	<code>EchoHorizonObstruction.lean</code>	done
External convergence inputs isolated with provenance	<code>ReggeConvergenceRegistry.lean</code>	done
<code>native_decide</code> basis disclosed	<code>Track1BCompilerTrustStatus.lean</code>	done
Corrected-Taylor all-cardinality reduction + algebraic lemma	<code>CorrectedTaylorHigherCardinality.lean</code>	done
Central contingency register, all eight modules wired	<code>QGScopeAudit.lean</code>	done
Pattern-A target type and justification	<code>ConditionalSlot.lean</code>	done
D2 reduced to a single named scalar-Dirichlet limit	<code>D2ScalarDirichletQuadratureLimit.lean</code> , <code>D2ScalarDirichletPartial.lean</code>	done
Repo-wide Pattern-A <code>ConditionalSlot</code> lift (48 shells)	scripted transform staged	open
Curvature-bearing scalar-Dirichlet limit; general triangulations; GHY boundary; TT/Lichnerowicz; Dirac algebra	named in <code>QGScopeAudit.lean</code>	open

7 Artifacts produced

New verified Lean modules (10). `QGScopeAudit`, `ReggeConvergenceRegistry`, `Track1BCompilerTrustStatus`, `D2ScalarDirichletQuadratureLimit`, `D2ScalarDirichletPartial`, `LedgerToGeometryBridge`, `EchoHorizonObstruction`, `CorrectedTaylorHigherCardinality`, `AdmissibleTriangulationProcedure`, `ConditionalSlot` (all under `IndisputableMonolith/Gravity/`).

Corrected Lean modules (8). `MasterTheoremUnconditional`, `DarkEnergyQGIntegration`, `BlackHoleEchoesFromBounce`, `BlackHoleEchoesSI`, `QGObservableSignalModels`, `QGChannelRungDerivation`, `DiscriminatorMatrix`, `Physics/QuantumGravityFromRS`.

Paper. `QG_Strong_Theory_Paper_20260611.tex`: abstract, echo section, falsifier matrix, strong-field table, master statement, status table, Bianchi scope, Taylor-gate status, and a new Contingency Register in the formalization appendix.

Tooling, plans, rules. `glm/qg_contingency_loop.py` (verified loop), `glm/pattern_a_conditionalslot_transf` (48-shell scanner), `plans/QG_Contingency_FixIt_Plan_20260621.html`, and an “autonomous loop operations” section added to the GLM operating rule (detachment via session leader, the `pkill -f` self-match pitfall, the maker/judge gate, reshape-don’t-rethrow).

8 What remains, stated plainly

The framework is now honestly scoped, with every contingency machine-visible. It is not a finished quantum-gravity theory, and the modules say so. The open items, each with a named owner in `QGScopeAudit.lean`:

1. The repo-wide `Pattern-A ConditionalSlot` lift (48 shells): the one can’t-skip structural item, staged with tooling and a worklist.
2. The curvature-bearing scalar-Dirichlet quadrature limit: `D2` is reduced to this single named real-sequence convergence; the limit itself is open.
3. General (non-product, non-flat) admissible triangulations; the Hartle–Sorkin/GHY boundary recovery; TT/Lichnerowicz operator convergence; and the Dirac constraint-algebra closure.
4. A horizon-consistent exterior echo mechanism, if one exists, to replace the rejected interior bounce.

The change in posture is the point. Before this session, the documents and some status records read as if the quantum-gravity theory were closed. After it, the genuinely proved core stands, every assumed bridge and external input is named and visible in the type system, the one physically wrong mechanism is proved impossible, and the remaining work is enumerated where a reader cannot miss it.